SEC P vs NP log 2026-05-12

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-18 13:47 UTC

SEC P vs NP — daily log 2026-05-12

52 cycles recorded

Free Cumulant Sum Bounded by Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Probability Theory × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-12 01:18 UTC
- **Entry ID**: `ecd8f062c47d`

Statement

For a |X|×|Y| matrix M defining a partial function f, let μ(M) = ∑_{k≥1} |c_k(M)| where c_k(M) are the free cumulants of M's entries. Then μ(M) ≤ O(√(R(f))) where R(f) is the randomized communication complexity of f. For DISJ_n, μ(DISJ_n) = Ω(n) holds.

Rationale

Free cumulants capture non-commutative independence structures in matrix entries, which may correlate with the information-theoretic cost of coordinating disjointness checks. The sum's growth rate could reflect the minimal communication required to resolve entangled dependencies.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38ecf086.py", line 75, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38ecf086.py", line 54, in run_trial
    cumulants = free_cumulants(M)
                ^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38ecf086.py", line 39, in free_cumulant
   R[i][j] -= sum(R[x][y] * R[i - x][j - y] for x in range(i) for y in range(j)) / (i * j)
           ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^~~~~~~~~
ZeroDivisionError: division by zero
```

Judge reasoning

Test crashed due to division by zero, preventing evaluation of conjecture | next: Fix implementation to handle zero division and retest with validated inputs

---

Plethysm Coefficient Gap in Permanent vs Determinant Decompositions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl Duality × Boolean Circuit Size for Permanent Polynomial
- **Recorded**: 2026-05-12 01:34 UTC
- **Entry ID**: `40591186a3ea`

Statement

For all $n \geq 2$, the plethysm coefficient λ(perm_n) of the symmetric power S^k(perm_n) exceeds λ(det_m) + $\varepsilon \cdot n^2$ for m < n^{1.5}, where $\varepsilon > 0$ is universal. This gap persists under linear substitutions preserving the polynomial's symmetry type.

Rationale

Plethysm coefficients encode symmetry constraints of polynomial representations. By linking them to circuit complexity, we may expose algebraic barriers to expressing permanent as a determinant-like circuit, leveraging representation-theoretic invariants resistant to natural proofs.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.19s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d9d81969.py", line 81, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d9d81969.py", line 58, in run_trial
    if lambda_perm_n > lambda_det_m + epsilon * n ** 2:
                       ~~~~~~~~~~~~~^~~~~~~~~~~~~~~~~~
OverflowError: int too large to convert to float
```

Judge reasoning

Test crashed with overflow error before producing data; cannot verify conjecture's validity | next: Implement arbitrary-precision arithmetic or numerical stability checks for large n

---

Symmetric Group Fourier Coefficients Inverse Proportional to ACC^0 Circuit Size for Sipser Function

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation Theory of Finite Groups × ACC^0 Circuit Size for Sipser Function
- **Recorded**: 2026-05-12 01:48 UTC
- **Entry ID**: `567baddf9ba5`

Statement

For the Sipser function S_n on n variables, the maximum absolute Fourier coefficient over the symmetric group S_n's irreducible representations is $\Theta(1/k)$, where k is the minimal ACC^0 circuit size computing S_n. This holds for all $n \geq 2$.

Rationale

Symmetric group representations capture combinatorial symmetries of permutation-based functions like Sipser's. Fourier coefficients over these representations quantify 'non-abelian harmonic analysis' of the function, potentially revealing structural obstructions to efficient ACC^0 computation.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_136fa469.py", line 80, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_136fa469.py", line 59, in run_trial
    for k in range(1, 2**n + 1):
                         ^
NameError: name 'n' is not defined
```

Judge reasoning

Test crashed due to undefined variable 'n' before producing results | next: Fix the NameError in test code by defining 'n' before usage

---

Integer Point Count in Solution Polytope Bounds Extended Frege Proof Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Geometry of Numbers × Extended Frege Proof Length
- **Recorded**: 2026-05-12 01:54 UTC
- **Entry ID**: `4798bdb0751b`

Statement

For a 3-SAT instance Φ with n variables, the number of integer solutions $S(\Phi)$ satisfies $S(\Phi) \geq$ 2^{n - $L(\Phi)$}, where $L(\Phi)$ is the length of the Extended Frege proof of Φ. Equality holds when Φ is a tautology with a unique solution.

Rationale

The geometry of numbers connects the combinatorial structure of integer solutions to algebraic constraints, while Extended Frege proofs formalize logical deductions. The conjecture posits that solution density directly impacts proof complexity via exponential trade-offs, exposing hidden geometric regularities in tautology refutations.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8072c074.py", line 82, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8072c074.py", line 64, in run_trial
    S_phi = sum(1 for _ in filter(dpll, [sat_formula]))
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8072c074.py", line 64, in <genexpr>
    S_phi = sum(1 for _ in filter(dpll, [sat_formula]))
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8072c074.py", line 27, in dpll
    literal, rest = sat_formula[0]
    ^^^^^^^^^^^^^
ValueError: too many values to unpack (expected 2)
```

Judge reasoning

Test crashed during execution, preventing data collection. The error suggests a structural issue with the sat_formula handling in the DPLL algorithm. | next: Debug the DPLL implementation's handling of sat_formula structure to resolve the unpacking error

---

Matroid Rank Bounded by Nisan-Wigderson Seed Length for 3-SAT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid Theory × Nisan-Wigderson PRG Seed Length
- **Recorded**: 2026-05-12 02:14 UTC
- **Entry ID**: `fa70962e79f5`

Statement

For any 3-SAT instance with n variables, the seed length of the Nisan-Wigderson PRG fooling it is at most the rank of the matroid formed by its clause hypergraph, where the rank is the maximal size of a set of clauses with pairwise disjoint variable support.

Rationale

Matroid rank captures combinatorial dependencies between clauses, which may constrain the pseudorandomness needed to fool the formula. This bridges structural properties of CNF formulas with PRG efficiency, leveraging matroid theory's role in encoding combinatorial constraints.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.01s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Run test with extended timeout and memory limits to complete execution

---

Matroid Rank Submodularity and Monotone Circuit Size for k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid Theory × Monotone Circuit Size for k-CLIQUE
- **Recorded**: 2026-05-12 02:20 UTC
- **Entry ID**: `b052cc185466`

Statement

For a k-CLIQUE CNF formula Φ on n variables, the rank function r(S) of the matroid M constructed from Φ satisfies r(S) ≥ Ω(n^{1/2}) for any subset S of size Ω(n^{1/2}), and the minimum number of elements needed to achieve r(S) = Ω(n^{1/2}) is Ω(n^{1/2}).

Rationale

The submodularity of the rank function in matroid theory could mirror the structure of monotone circuits, where dependencies between clauses (elements) affect the circuit's size. By linking the rank's growth rate to the circuit's complexity, we might uncover structural constraints.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Test crashed before producing data, preventing evaluation of support fraction or counterexamples. | next: Re-run test with detailed logging to identify crash cause and collect empirical evidence

---

Symplectic Rank of Disjointness Communication Matrix Bounds Randomized Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Symplectic Geometry × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-12 02:46 UTC
- **Entry ID**: `8acf355ada09`

Statement

The symplectic rank of the DISJOINTNESS communication matrix M is Ω(n), where the symplectic rank is defined as the minimal k such that M decomposes into k rank-1 symplectic tensors under a fixed symplectic form. This lower bounds the randomized communication complexity of DISJOINTNESS.

Rationale

Symplectic geometry provides invariants that capture structural properties of matrices, which may correlate with communication complexity. The symplectic rank, as a decomposition invariant, could expose hidden algebraic constraints in the matrix that mirror information-theoretic barriers.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dfb495a0.py", line 102, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dfb495a0.py", line 74, in run_trial
    if not is_symplectic_form(M, n):
           ^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dfb495a0.py", line 44, in is_symplectic
   return matrix_multiply(matrix_multiply(M, M), I) == -I and matrix_multiply(transpose(M), M) == -
I
                                                         ^^
TypeError: bad operand type for unary -: 'list'
```

Judge reasoning

Test crashed due to type error in matrix operations, preventing evaluation of conjecture | next: Fix matrix arithmetic implementation to handle symplectic form verification correctly

---

Schur Coefficient Density Inverse Proportional to Frege Proof Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Symmetric Function Theory × Frege Proof Length
- **Recorded**: 2026-05-12 03:10 UTC
- **Entry ID**: `2b3f03ad1b52`

Statement

For a homogeneous polynomial f of degree d in n variables, let S(f) denote the number of non-zero coefficients in its Schur decomposition. Then, the Frege proof length of f = 0 is at least Ω(S(f)^{1/2})

Rationale

The density of Schur coefficients reflects the complexity of the polynomial's representation, which might correlate with the difficulty of proving its identity through Frege systems. Symmetric function theory provides a framework to decompose polynomials into irreducible components, exposing structural properties that could influence proof complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
ue': 1, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': ''}
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
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TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_98998d42.py", line 121, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_98998d42.py", line 121, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                                   ~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

Test crashed due to KeyError before producing reliable data | next: Fix the test code to handle missing 'seed' field and re-run with proper error handling

---

Free Cumulant Sum Bounded by Log-Size for Read-Twice BPs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Probability Theory × Read-Twice Branching Programs
- **Recorded**: 2026-05-12 03:25 UTC
- **Entry ID**: `93b47d86db79`

Statement

For any read-twice BP P computing IP_2, the sum of free cumulants $\Sigma_{k=1}^{\infty} \kappa_k(P) \leq \log(\text{size}(P)) + O(1)$. For random read-twice BPs over GF(2), $\Sigma \kappa_k(P) = \Theta(\log n)$.

Rationale

Free cumulants capture non-commutative independence structures; their sum could distinguish read-twice BPs (with limited independence) from unrestricted BPs (with full independence). IP_2's inherent symmetry might force linear cumulant growth.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.01s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out without producing supporting data or counterexamples | next: Increase timeout duration and re-run tests with extended computation limits

---

Condition Number of Moment Matrix Inversely Proportional to SOS Degree for Max-CUT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Convex Geometry (Condition Number of Moment Matrices) × SOS Degree for Max-CUT
- **Recorded**: 2026-05-12 04:34 UTC
- **Entry ID**: `f89319bf92ad`

Statement

For a random max-CUT instance with n variables, the condition number κ of the moment matrix in the degree-d SOS relaxation satisfies $\kappa = \Theta(1/d^2)$, and the SOS degree required to achieve an approximation ratio of 0.878 - ε is $\Omega(\sqrt{(d)})$

Rationale

The condition number of the moment matrix reflects the sensitivity of the SOS relaxation to perturbations. A lower condition number implies better-conditioned matrices, which may require lower SOS degrees to achieve the same approximation ratio. This link could reveal geometric properties of the feasible region in the SOS hierarchy

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cb36e3f3.py", line 92, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cb36e3f3.py", line 65, in run_trial
    inv_M = matrix_inv(M)
            ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cb36e3f3.py", line 42, in matrix_inv
    A_rref = gaussian_elimination(A_augmented)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cb36e3f3.py", line 28, in gaussian_elim
    raise ValueError("Matrix is singular")
ValueError: Matrix is singular
```

Judge reasoning

Test crashed with singular matrix error, preventing validation of conjecture | next: Investigate why matrix is singular for tested parameters and verify matrix construction

---

Projective Plane Incidence Bounds Nisan-Wigderson Seed Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Finite Geometry (Projective Planes) × Nisan-Wigderson PRG Seed Length
- **Recorded**: 2026-05-12 04:40 UTC
- **Entry ID**: `c9054923e964`

Statement

For a CNF Φ with n variables whose clause incidence matrix is a projective plane of order q, the Nisan-Wigderson seed length for Φ is $\leq q^2 + q + 1$. Conversely, if Φ has seed length $> q^2 + q + 1$, its incidence matrix cannot be a projective plane of order q.

Rationale

Projective planes exhibit optimal combinatorial expansion properties that could constrain pseudorandom generator seed requirements. The bound q^2+q+1 matches the number of lines in the plane, suggesting a direct geometric constraint on the PRG's pseudorandomness threshold.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_54288a4f.py", line 148, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_54288a4f.py", line 110, in run_trial
    cnf = generate_cnf_from_plane(points, lines)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_54288a4f.py", line 85, in generate_cnf_
    clause = [j + 1 for j in range(m) if (i // q, i % q) in lines[j]]
                                               ^
NameError: name 'q' is not defined
```

Judge reasoning

Test crashed due to undefined variable ‘q’ in the generation logic, preventing validation of the conjecture | next: Fix the code to properly define ‘q’ in the projective plane generation and re-run tests

---

Fourier Coefficient Spread and ACC^0 Circuit Lower Bounds

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Fourier Analysis on Finite Groups × ACC^0 Circuit Size
- **Recorded**: 2026-05-12 05:17 UTC
- **Entry ID**: `90abe309451a`

Statement

For any boolean function f:{0,1}^n → {0,1}, if the sum of absolute Fourier coefficients ∑|$\hat{f}$(S)| ≥ 2^{n/2} √n, then f cannot be computed by an ACC^0 circuit of size 2^{n/2}.

Rationale

The spread of Fourier coefficients reflects correlation with parity functions. ACC^0 circuits, limited to bounded-depth polynomial-size gates, struggle to capture such correlations, making large Fourier sums a barrier to efficient computation.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.07s`

```
tric_name': 'Fourier Coefficient Sum', 'metric_value': 0.64453125, 'instances_tested': 1, 'conjectur
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.626953125, 'instances_tested': 1
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.59765625, 'instances_tested': 1,
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.5625, 'instances_tested': 1, 'co
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.634765625, 'instances_tested': 1
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.625, 'instances_tested': 1, 'con
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.546875, 'instances_tested': 1, '
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.56640625, 'instances_tested': 1,
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.5546875, 'instances_tested': 1,
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.59765625, 'instances_tested': 1,
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.658203125, 'instances_tested': 1
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.625, 'instances_tested': 1, 'con
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.666015625, 'instances_tested': 1
RESULT: SUPPORTED mean=0.6087239583333334 std=0.03415580763030604 support_fraction=1.0
```

Judge reasoning

The test used n=1 where the conjecture’s threshold (2^{n/2}√n) is ~1.414, but all metric values (<1) never reached this threshold. Thus, the conjecture’s condition was never tested. | next: Test with n≥2 where 2^{n/2}√n ≥2 to validate the conjecture’s actual condition

---

Fourier Coefficient Decay Bounds AC^0 Circuit Size for Parity-Insensitive Functions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Fourier Analysis on Finite Groups × AC^0 Circuit Size
- **Recorded**: 2026-05-12 05:43 UTC
- **Entry ID**: `29d6f7d21e9f`

Statement

For any parity-insensitive Boolean function f: {0,1}^n → {0,1}, the sum of absolute Fourier coefficients ∑_{S≠∅} |f̂(S)| decays exponentially with n iff the minimal AC^0 circuit size computing f is Ω(2^{n/2}).

Rationale

Parity-insensitive functions exhibit Fourier coefficient concentration patterns that may correlate with algebraic complexity. The exponential decay threshold could expose a structural gap between Fourier analytic simplicity and circuit complexity, leveraging the inherent symmetry of AC^0 computation.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.29s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0b03edf7.py", line 97, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0b03edf7.py", line 67, in run_trial
    f_hat = compute_fourier_coefficients(clauses, n)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0b03edf7.py", line 38, in compute_fouri
    return [abs(coeff / (1 << n)) for coeff in fast_walsh_hadamard_transform(f)]
                                               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0b03edf7.py", line 23, in fast_walsh_ha
    f[j] += f[j + n // 2]
            ~^^^^^^^^^^^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError before producing results, preventing evaluation of conjecture validity | next: Debug fast_walsh_hadamard_transform implementation to fix index out of range error

---

Invariant Ring Degree Bounds SOS Refutation Degree for Symmetric CSPs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Invariant Theory × SOS Refutation Degree for CSPs
- **Recorded**: 2026-05-12 06:36 UTC

- **Entry ID**: `3a288632d238`

Statement

For a symmetric CSP instance Φ invariant under a finite group G, the minimal degree d of a generating set of the invariant ring of G is inversely proportional to the SOS refutation degree of Φ. Specifically, the SOS refutation degree is at least Ω(1/d).

Rationale

Symmetric CSPs can be analyzed via their invariant rings, which capture the group's action. The degree of the generators may limit the SOS hierarchy's ability to refute the instance, as higher-degree generators might require higher SOS levels.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9d483de9.py", line 74, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9d483de9.py", line 51, in run_trial
    instance = generate_symmetric_csp(3)  # Example: n=3
               ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9d483de9.py", line 37, in generate_symm
    while not is_symmetric(instance):
              ^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9d483de9.py", line 30, in is_symmetric
    permuted_instance = [instance[g[i]] for i in range(len(instance))]
                                  ~^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError before producing results, preventing evaluation of conjecture validity. | next: Fix the index out-of-range error in generate_symmetric_csp() and re-run tests with multiple seeds

---

Noncommutative L^p Norm of Communication Matrix Bounds Randomized Disjointness Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Noncommutative L^p Geometry × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-12 06:43 UTC
- **Entry ID**: `cfd6712a7d9a`

Statement

For a random instance M of DISJOINTNESS on n elements, the noncommutative L^p norm ||M||_{p} ≥ Ω(√n) for p=2, with equality achieved when M is a balanced bipartite graph with uniform edge weights.

Rationale

Noncommutative L^p norms capture operator-theoretic structure of tensors, which may encode entanglement-like constraints in communication protocols. The L^2 norm's relation to matrix rank could expose hidden geometric barriers in randomized protocols for DISJOINTNESS.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.10s`


```
c_name': 'Frobenius Norm', 'metric_value': 28.284271247461902, 'instances_tested': 30, 'conjecture_h
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 28.284271247461902, 'instances_tested': 30,
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 28.284271247461902, 'instances_tested': 30,
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TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 28.284271247461902, 'instances_tested': 30,
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 28.284271247461902, 'instances_tested': 30,
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 28.284271247461902, 'instances_tested': 30,
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 28.284271247461902, 'instances_tested': 30,
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 28.284271247461902, 'instances_tested': 30,
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 28.284271247461902, 'instances_tested': 30,
RESULT: SUPPORTED mean=28.284271247461902 std=undefined support_fraction=1.0
```


Judge reasoning

Safety rail: critic_challenge | original judge: | next: NONE

Matroid Rank Inverse Proportional to Monotone DNF Size for k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid Theory × Monotone Circuit Size for k-CLIQUE
- **Recorded**: 2026-05-12 07:48 UTC
- **Entry ID**: `afe153586854`

Statement

For a k-CLIQUE instance on n vertices, let r be the maximum number of pairwise disjoint k-cliques. Then, the minimal size of a monotone DNF formula computing the k-CLIQUE function is Ω(n^{2/r}).

Rationale

The matroid rank captures the structural decomposition of the graph into disjoint cliques, and higher ranks imply that the graph can be decomposed into more cliques, which might allow for a more efficient DNF representation. Conversely, lower ranks (indicating fewer decompositions) would require larger DNFs.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.03s`

```
ed': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'dnf_size', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': Tru
TRIAL: {'metric_name': 'dnf_size', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': Tru
TRIAL: {'metric_name': 'dnf_size', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': Tru
TRIAL: {'metric_name': 'dnf_size', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': Tru
TRIAL: {'metric_name': 'dnf_size', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': Tru
TRIAL: {'metric_name': 'dnf_size', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': Tru
TRIAL: {'metric_name': 'dnf_size', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': Tru
TRIAL: {'metric_name': 'dnf_size', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': Tru
TRIAL: {'metric_name': 'dnf_size', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': Tru
TRIAL: {'metric_name': 'dnf_size', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': Tru
TRIAL: {'metric_name': 'dnf_size', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': Tru
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TRIAL: {'metric_name': 'dnf_size', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': Tru
TRIAL: {'metric_name': 'dnf_size', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': Tru
TRIAL: {'metric_name': 'dnf_size', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': Tru
RESULT: SUPPORTED mean=40.00 std=0 support_fraction=1
```

Judge reasoning

Single instance tested with n=1 makes Ω(n^{2/r}) bound vacuous. Support fraction (1) meets threshold but insufficient data points violate pre-registered criterion. | next: Test with multiple instances across varying n and r values to validate scaling behavior

---

Matroid Rank Submodularity in Monotone DNF for k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid Theory (Polymatroid Expansion) × Monotone Circuit Size for k-CLIQUE
- **Recorded**: 2026-05-12 07:54 UTC
- **Entry ID**: `16b4e4e9c316`

Statement

Let μ(F) be the rank of the transversal matroid induced by a monotone DNF formula F. For all F with size poly(n), μ(F) = O(log n). For k-CLIQUE instances, μ(F) = Ω(n^{1 - 1/k})

Rationale

Transversal matroid rank captures submodularity of DNF terms, aligning with monotone circuit structure. k-CLIQUE's combinatorial constraints force high rank due to clique edge intersections, creating a gap between general DNFs and clique-specific formulas.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c7104728.py", line 101, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c7104728.py", line 63, in run_trial
    dnf = generate_dnf(n)
          ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c7104728.py", line 24, in generate_dnf
    dnf.add(tuple(sorted(term) for term in terms))
TypeError: unhashable type: 'list'
```

Judge reasoning

Test crashed due to unhashable type error, preventing data collection to evaluate conjecture | next: Fix generate_dnf() to handle hashable types and re-run experiments

---

Tensor Product Multiplicity Inverse Proportional to Monotone Circuit Size for Permanent

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation Theory of Symmetric Groups (Tensor Product Decompositions) × Monotone Circuit Size for Permanent
- **Recorded**: 2026-05-12 08:01 UTC
- **Entry ID**: `7a5d68832c2b`

Statement

For a random 3-regular hypergraph H_n on n vertices, let m(λ) denote the multiplicity of irreducible representation λ in the decomposition of V_{n} ⊗ V_{n} under the symmetric group S_n. Then m(λ) = Θ(1/size_{mono}(Perm_n)) where size_{mono}(Perm_n) is the minimal monotone circuit size computing the permanent on n variables.

Rationale

Tensor product multiplicities encode combinatorial symmetries of permutation matrices, which are structurally similar to the monotone circuits required to compute permanent. The inverse proportionality suggests that richer representation structures correlate with lower circuit complexity, exposing hidden algebraic constraints.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_efe16db4.py", line 61, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_efe16db4.py", line 40, in run_trial
    mλ = hook_length_formula(n, λ)
         ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_efe16db4.py", line 30, in hook_length_f
    hook_lengths = [[n - row - col + 1 for col in range(col, n)] for row in range(row, n)]
                                                                                   ^^^
NameError: name 'row' is not defined. Did you mean: 'pow'?
```

Judge reasoning

Test crashed due to NameError in hook_length_formula implementation, preventing data collection | next: Fix the hook_length_formula implementation to properly define row/col indices and re-run trials

---

Persistent Homology of Read-Twice BPs Bounded by Log-Size

- **Verdict**: INCONCLUSIVE
- **Bridge**: Persistent Homology × Read-Twice Branching Programs
- **Recorded**: 2026-05-12 08:14 UTC
- **Entry ID**: b4f04707e5bb

Statement

For a read-twice BP P computing a function f: {0,1}^n → {0,1}, let β(P) = ∑_{k} (dim H_k) * persistence_k, where H_k is the k-th homology group of P's computation graph. Then β(P) = O(log size(P)), but β(IP_2) = Ω(n^2) where IP_2 is the inner product mod 2 function.

Rationale

Persistent homology captures hierarchical topological features of computation graphs. Read-once BPs have acyclic structures limiting homology complexity, while read-twice BPs may encode richer topological patterns. IP_2's exponential BP complexity creates persistent high-dimensional holes.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 1.05s


```
c_value': 1, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
```

```
TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
TRIAL: {'metric_name': 'beta(P)', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True
RESULT: SUPPORTED mean=1.0 std=0.0 support_fraction=1.0
```

Judge reasoning

All 10 trials used the same single instance, violating the n-too-small failure mode. The support_fraction=1.0 is statistically vacuous for scaling claims. | next: Test with ≥80 diverse instances across varying n to validate scaling behavior

---

Hilbert Function Growth Inversely Proportional to Extended Frege Proof Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Geometry (Hilbert Functions) × Extended Frege Proof Length
- **Recorded**: 2026-05-12 08:22 UTC
- **Entry ID**: `f78ced0ca719`

Statement

For a CNF tautology Φ with n variables, the Hilbert function H(ideal(Φ), d) at degree d = ⌈n/2⌉ satisfies H(ideal(Φ), d) = Θ(2^{-λ(n)}), where λ(n) is the Extended Frege proof length of Φ.

Rationale

Hilbert functions capture the complexity of polynomial ideals, which may encode logical dependencies in proofs. Inverse proportionality suggests deeper algebraic structure in proof systems, potentially revealing connections between algebraic geometry and proof complexity barriers.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_906170a2.py", line 190, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_906170a2.py", line 156, in run_trial
    ideal = groebner_basis(phi, 2**32 - 5)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_906170a2.py", line 113, in groebner_bas
   s = subtract_matrices(basis[i], multiply_matrices([[basis[j][k] * mod_inverse(basis[i][j], mod)]]
```

```
                                                                                                            ^
NameError: name 'k' is not defined
```

Judge reasoning

Test crashed due to NameError, preventing data collection. Support condition cannot be evaluated without results. | next: Fix the NameError in matrix operations by defining variable 'k' and revalidate the test implementation

---

Block Design Discrepancy and Communication Complexity Lower Bound

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorial Design Theory × Communication Complexity (Discrepancy of Set Disjointness)
- **Recorded**: 2026-05-12 08:32 UTC
- **Entry ID**: `8a6982e09855`

Statement

For a symmetric (v, k, λ) block design, the discrepancy of the communication matrix for the set disjointness function is Ω(√(λ v / k))

Rationale

Block designs impose regular structure on incidence relations, which may constrain the discrepancy of associated communication matrices, offering a combinatorial lens to bound communication complexity

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.03s`

```
etric_value': None, 'instances_tested': 0, 'conjecture_holds': False, 'counterexample': 'design_not_
TRIAL: {'metric_name': 'discrepancy', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'discrepancy', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'discrepancy', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'discrepancy', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'discrepancy', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'discrepancy', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'discrepancy', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'discrepancy', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'discrepancy', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'discrepancy', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'discrepancy', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'discrepancy', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
RESULT: FALSIFIED counterexample="design_not_possible" first_failing_seed=11
```

Judge reasoning

No empirical tests were executed (instances_tested=0), and the counterexample is a logical contradiction rather than an empirical failure. | next: Implement tests with concrete symmetric block design parameters and measure discrepancy values directly

---

Second Eigenvalue Inverse Proportional to Resolution Proof Length for Tseitin Formulas

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Spectral Graph Theory × Resolution Proof Length for Tseitin Formulas
- **Recorded**: 2026-05-12 08:44 UTC
- **Entry ID**: `67d14839d492`

Statement

For a random Tseitin formula derived from a d-regular expander graph with n nodes, the resolution proof length is $\Theta(1 / \lambda_2)$, where λ_2 is the second largest eigenvalue of the adjacency matrix. For all such instances with $\lambda_2 \leq 1/n$, the proof length exceeds n^{1.5}.

Rationale

Expander graphs' spectral properties govern their resistance to local reasoning. The second eigenvalue λ_2 quantifies the graph's expansion, and its inverse may correlate with the difficulty of deriving contradictions via resolution, as expanders require global interactions.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3aa44c0b.py", line 147, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3aa44c0b.py", line 101, in run_trial
    lambda_2 = eigenvalues(G)[1]
               ^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3aa44c0b.py", line 59, in eigenvalues
    y = matrix_multiplication(A, x)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3aa44c0b.py", line 24, in matrix_multip
    C[i][j] += A[i][k] * B[k][j]
                         ~~~~^^^
TypeError: 'float' object is not subscriptable
```

Judge reasoning

Test crashed due to type error in matrix multiplication, preventing data collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Fix matrix multiplication to handle vectors vs matrices correctly and retest with validated input types

---

Real Radical Dimension Bounds SOS Degree for Max-CUT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × SOS Degree for Max-CUT
- **Recorded**: 2026-05-12 09:26 UTC
- **Entry ID**: `02160132106f`

Statement

For a Max-CUT instance with n variables, the SOS degree required to approximate the maximum cut is at least the dimension of the real radical of the ideal generated by the constraint polynomials. Formally, if I is the ideal generated by {x_i^2 - x_i | 1 ≤ i ≤ n} ∪ {∑_{(i,j)∈E} (1 - x_i - x_j + x_i x_j) - α}, then dim(√I) ≥ d implies SOS degree ≥ d.

Rationale

The real radical captures the semialgebraic structure of the feasible region for Max-CUT. Higher-dimensional varieties may require higher-degree SOS certificates to capture their geometry, linking algebraic complexity to proof complexity. This conjecture bridges real algebraic geometry's intrinsic dimensionality with SOS hierarchy's computational requirements.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.03s`

```
'counterexample': ''}
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': T
RESULT: SUPPORTED mean=20.0 std=0.0 support_fraction=1.0
```

Judge reasoning

parse_fail | next: NONE

Noncommutative Fourier Coefficient Sum Inversely Proportional to ACC⁰ Circuit Size for Sipser Functions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Noncommutative Harmonic Analysis × ACC^0 Circuit Size for Sipser Function
- **Recorded**: 2026-05-12 10:22 UTC
- **Entry ID**: `4d29c6985de2`

Statement

For a Boolean function f on n variables, the sum of absolute values of its noncommutative Fourier coefficients on the symmetric group S_3 is inversely proportional to the ACC^0 circuit size of f, specifically, $\Sigma |F(k)| = \Theta(1 / size\_\{ACC^0\}(f))$

Rationale

Noncommutative harmonic analysis reveals structural properties of functions on non-abelian groups, which may be inherently complex for ACC^0 circuits to replicate, offering a new angle for lower bound proofs

Novelty

- Judge: `INCONCLUSIVE` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3e221ce4.py", line 62, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3e221ce4.py", line 46, in run_trial
    f_g = group_action(f, g)
          ^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3e221ce4.py", line 27, in group_action
    return [f(tuple(g[i] for i in range(len(f)))) for f in f]
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
TypeError: 'function' object is not iterable
```

Judge reasoning

Test crashed due to TypeError, preventing data collection. Support fraction cannot be evaluated. | next: Fix the 'function object not iterable' error in group_action() by verifying input types

---

SOS Degree Lower Bound via Eigenvalue Count in Moment Matrix

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Random Matrix Theory × SOS Degree for Max-CUT
- **Recorded**: 2026-05-12 10:30 UTC
- **Entry ID**: `929f2e4b21a8`

Statement

For a random Max-CUT instance with n variables, the minimal SOS degree required to refute it is at least $\log_2(k)$, where k is the number of non-zero eigenvalues of the corresponding moment matrix.

Rationale

The structure of the moment matrix's eigenvalues reflects the complexity of the polynomial constraints, and thus the SOS degree needed to capture these constraints is logarithmic in the number of non-zero eigenvalues.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: `0.11s`

```
ture_holds": True, "counterexample": ""}
TRIAL: {"seed": 71, "metric_name": "SOS Degree", "metric_value": 5, "instances_tested": 1, "conjectur
TRIAL: {"seed": 89, "metric_name": "SOS Degree", "metric_value": 4, "instances_tested": 1, "conjectur
TRIAL: {"seed": 103, "metric_name": "SOS Degree", "metric_value": 6, "instances_tested": 1, "conjectu
TRIAL: {"seed": 127, "metric_name": "SOS Degree", "metric_value": 3, "instances_tested": 1, "conjectu
TRIAL: {"seed": 149, "metric_name": "SOS Degree", "metric_value": 4, "instances_tested": 1, "conjectu
TRIAL: {"seed": 167, "metric_name": "SOS Degree", "metric_value": 5, "instances_tested": 1, "conjectu
TRIAL: {"seed": 191, "metric_name": "SOS Degree", "metric_value": 6, "instances_tested": 1, "conjectu
TRIAL: {"seed": 211, "metric_name": "SOS Degree", "metric_value": 5, "instances_tested": 1, "conjectu
TRIAL: {"seed": 233, "metric_name": "SOS Degree", "metric_value": 4, "instances_tested": 1, "conjectu
TRIAL: {"seed": 257, "metric_name": "SOS Degree", "metric_value": 5, "instances_tested": 1, "conjectu

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fc92df40.py", line 88, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fc92df40.py", line 62, in run_trial
    eigenvals = eigenvalues(G)
                ^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fc92df40.py", line 43, in eigenvalues
    factor = 1 / matrix[i][i]
             ~~^~~~~~~~~~~~~~
ZeroDivisionError: float division by zero
```

Judge reasoning

Test crashed with ZeroDivisionError, preventing full validation of the conjecture's support fraction. | next: Re-run tests with fixed division-by-zero handling and verify support_fraction >= 0.8

Kronecker Coefficient Gap in Symmetric Decompositions of Permanent Polynomials

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation Theory of Symmetric Groups (Kronecker coefficients) × Boolean Circuit Size for Permanent Polynomial

- **Recorded**: 2026-05-12 11:18 UTC
- **Entry ID**: `b1e1fdcf3c47`

Statement

For a random n×n matrix M with entries ±1, the Kronecker coefficient g(λ, μ, ν) for the decomposition of Sym^2(Perm_n) ⊗ Sym^2(Perm_n) is Ω(n^2) and O(n^3), where λ, μ, ν are partitions of 2n. The ratio g(λ, μ, ν)/n^2 is strictly greater than 1 for all n ≤ 40.

Rationale

Kronecker coefficients encode the multiplicity of irreducible representations in tensor products, which could reflect the inherent symmetry complexity of permanent polynomials. Their growth rate may correlate with the minimal circuit size required to compute the permanent, as symmetric decompositions often require more terms than antisymmetric ones.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.15s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c507cd2f.py", line 97, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c507cd2f.py", line 81, in run_trial
    ratio = coeff / (n**2)
            ~~~~~~^~~~~~~~
OverflowError: integer division result too large for a float
```

Judge reasoning

Test crashed with overflow error during ratio calculation, preventing verification of conjecture for n ≤ 40 | next: Implement arbitrary-precision arithmetic or reformulate ratio calculation to avoid overflow

---

Schur-Weyl Decomposition Component Count vs. Monotone Circuit Size for Permanent-like CNFs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl Duality × Monotone Circuit Size
- **Recorded**: 2026-05-12 11:48 UTC
- **Entry ID**: `965ac372534a`

Statement

For any CNF Φ with n variables and m clauses, let C(Φ) denote the number of irreducible components in the Schur-Weyl decomposition of the tensor product of its clause indicator vectors. Then, for all unsatisfiable Φ with m = Θ(n^2), C(Φ) ≥ 2^{n/2} / poly(n) and the monotone circuit size of Φ is Ω(C(Φ) * log n).

Rationale

Schur-Weyl duality decomposes tensor products into symmetric/antisymmetric subspaces, revealing structural complexity. Permanent-like CNFs exhibit richer decomposition patterns due to their symmetric nature, which may correlate with inherent monotone circuit lower bounds via representation-theoretic obstructions.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `2.43s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_754b3988.py", line 88, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_754b3988.py", line 64, in run_trial
    tensor_product = [sum(v[i] * w[i] for v, w in zip(vec1, vec2)) for vec1, vec2 in zip(*clause_indicat
                                                                              ^^^^^^^^^^
ValueError: too many values to unpack (expected 2)
```

Judge reasoning

Test crashed during tensor product computation, preventing data collection. The ValueError indicates a critical error in the implementation that needs debugging. | next: Fix the unpacking error in the tensor product calculation and revalidate the test setup

---

Real Radical Degree Lower Bound for AC^0 Circuits Computing PARITY

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × AC^0 Circuit Size for PARITY
- **Recorded**: 2026-05-12 12:40 UTC
- **Entry ID**: `6554531eadaa`

Statement

For any AC^0 circuit C computing PARITY on n inputs, the real radical of the ideal generated by the polynomial equations defining C has degree Ω(2^{n/2})

Rationale

The real radical captures the essential structure of the polynomial equations, and its degree reflects the complexity required to represent PARITY, which AC^0 circuits cannot efficiently compute

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.13s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_66b6d873.py", line 70, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_66b6d873.py", line 47, in run_trial
    circuit = generate_ac0_circuit(n)
              ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_66b6d873.py", line 26, in generate_ac0_
    left = generate_ac0_circuit(n // 2)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_66b6d873.py", line 26, in generate_ac0_
    left = generate_ac0_circuit(n // 2)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_66b6d873.py", line 26, in generate_ac0_
    left = generate_ac0_circuit(n // 2)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  [Previous line repeated 1 more time]
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_66b6d873.py", line 28, in generate_ac0_
    return [left[i] ^ right[i % len(right)] for i in range(n)]
            ~~~~^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed due to infinite recursion and IndexError, preventing data collection. The error suggests a flaw in the circuit generation code rather than directly refuting the conjecture. | next: Fix the generate_ac0_circuit function to handle base cases (e.g., n=0) and test with small n values to resolve recursion errors

---

Noncommutative Fourier Coefficient Spread and Read-Twice BP Size

- **Verdict**: INCONCLUSIVE
- **Bridge**: Noncommutative Harmonic Analysis × Read-Twice Branching Programs for IP_2
- **Recorded**: 2026-05-12 13:19 UTC
- **Entry ID**: 1434c5493146

Statement

For a read-twice BP P computing IP_2, let Φ(P) denote its noncommutative Fourier transform. The spread of Φ(P)'s coefficients, defined as max_k |Φ(P)[k]| - min_k |Φ(P)[k]|, is Ω(n) for the trivial BP computing IP_2, but O(log size(P)) for all other read-twice BPs.

Rationale

Noncommutative Fourier analysis captures BP structure via operator-valued transforms, and coefficient spread reflects complexity, distinguishing IP_2 from other BPs via scale-sensitive invariants.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `180.36s`

```
_tested': 1, 'conjecture_holds': False, 'counterexample': ''}
TRIAL: {'metric_name': 'coefficient_spread', 'metric_value': 8.860411591808158e+38, 'instances_teste
TRIAL: {'metric_name': 'coefficient_spread', 'metric_value': 2.5377756348771555e+30, 'instances_test
TRIAL: {'metric_name': 'coefficient_spread', 'metric_value': 2.4148559913460305e+22, 'instances_test
TRIAL: {'metric_name': 'coefficient_spread', 'metric_value': 788171.4782896783, 'instances_tested':
TRIAL: {'metric_name': 'coefficient_spread', 'metric_value': 1.7846887383448086e+51, 'instances_test
TRIAL: {'metric_name': 'coefficient_spread', 'metric_value': 1.0341559093873158e+46, 'instances_test
TRIAL: {'metric_name': 'coefficient_spread', 'metric_value': 1982897212.6256335, 'instances_tested'
TRIAL: {'metric_name': 'coefficient_spread', 'metric_value': 2.8758744113973884e+92, 'instances_test
TRIAL: {'metric_name': 'coefficient_spread', 'metric_value': 3.450988614046317e+25, 'instances_teste
TRIAL: {'metric_name': 'coefficient_spread', 'metric_value': 2.1992540262651752e+86, 'instances_test
TRIAL: {'metric_name': 'coefficient_spread', 'metric_value': 1.0928913759696957e+34, 'instances_test
TRIAL: {'metric_name': 'coefficient_spread', 'metric_value': 4.4663753812146135e+28, 'instances_test
RESULT: FALSIFIED counterexample="trivial" first_failing_seed=11
```

Judge reasoning

Safety rail: critic_challenge_falsified | original: The conjecture is refuted by a counterexample where the coefficient spread for the trivial BP is significantly lower than Ω(n) across multiple trials. | next: Analyze the trivial BP's Fourier transform structure to understand why its coefficient spread violates the conjectured Ω(n) lower bound.

Invariant Ring Generator Count Gap in Read-Twice BPs for IP_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Invariant Theory × Read-Twice Branching Programs
- **Recorded**: 2026-05-12 13:27 UTC
- **Entry ID**: `5c395af825ac`

Statement

For a read-twice BP P computing a function f, the number of generators of the invariant ring of f is O(log size(P)), but for the inner product mod 2 function (IP_2), this count is Ω(n).

Rationale

Invariant rings capture symmetries of polynomial representations; read-twice BPs may exploit such structures efficiently, while IP_2's inherent symmetry complexity forces exponential growth in generator counts for read-once BPs.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.03s`

```
TRIAL: {"seed": 11, **result}
TRIAL: {"seed": 23, **result}
TRIAL: {"seed": 37, **result}
TRIAL: {"seed": 53, **result}
```

```
TRIAL: {"seed": 71, **result}
TRIAL: {"seed": 89, **result}
TRIAL: {"seed": 103, **result}
TRIAL: {"seed": 127, **result}
TRIAL: {"seed": 149, **result}
TRIAL: {"seed": 167, **result}
TRIAL: {"seed": 191, **result}
TRIAL: {"seed": 211, **result}
TRIAL: {"seed": 233, **result}
TRIAL: {"seed": 257, **result}
TRIAL: {"seed": 277, **result}
TRIAL: {"seed": 311, **result}
TRIAL: {"seed": 347, **result}
TRIAL: {"seed": 389, **result}
TRIAL: {"seed": 421, **result}
TRIAL: {"seed": 463, **result}
TRIAL: {"seed": 503, **result}
TRIAL: {"seed": 547, **result}
TRIAL: {"seed": 593, **result}
TRIAL: {"seed": 631, **result}
TRIAL: {"seed": 677, **result}
TRIAL: {"seed": 727, **result}
TRIAL: {"seed": 773, **result}
TRIAL: {"seed": 821, **result}
TRIAL: {"seed": 877, **result}
TRIAL: {"seed": 929, **result}
RESULT: INCONCLUSIVE insufficient data or too many failures
```

Judge reasoning

Test results show insufficient data to meet the 80% support threshold; no counterexample found but inconclusive outcome reported. | next: Run additional trials with larger n and more seeds to improve statistical confidence

---

Noncommutative Fourier Coefficient Sum Bounded by Log-Size for Read-Twice BPs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Noncommutative Harmonic Analysis (Fourier analysis on symmetric groups) × Read-Twice Branching Programs for IP_2
- **Recorded**: 2026-05-12 14:34 UTC
- **Entry ID**: `f273f29032d3`

Statement

For any read-twice branching program P computing a function f: {0,1}^n → {0,1}, the sum of absolute values of noncommutative Fourier coefficients of f over the symmetric group S_n is O(log n). In contrast, for the inner product mod 2 function IP_2, this sum is Ω(n^2).

Rationale

Noncommutative harmonic analysis on symmetric groups captures permutation symmetries, which are more structured in read-twice BPs. The IP_2 function's high symmetry leads to large Fourier coefficients, while read-twice BPs have more constrained structures.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `241.62s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out without producing data; cannot verify support fraction or counterexamples | next: Increase timeout duration and re-run test with extended computation limits

---

Schur-Weyl Multiplicity Gap in Symmetric Powers of Permanent vs Determinant

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl Duality × Boolean Circuit Size for Permanent Polynomial
- **Recorded**: 2026-05-12 15:07 UTC
- **Entry ID**: `056a10374551`

Statement

For all n ≤ 40, the sum of squares of Kronecker coefficients in the decomposition of S^k(perm_n) exceeds that of S^k(det_m) by Ω(2^{n/2}) for m < n^{1.5}, and this gap persists under linear substitutions.

Rationale

Schur-Weyl duality decomposes symmetric powers into irreducible representations, revealing structural differences between permanent and determinant. The Kronecker coefficients encode combinatorial complexity, potentially linking representation-theoretic gaps to circuit lower bounds.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.00s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing verification of conjecture. | next: Increase time limit and re-run for n ≤ 20 with optimized code

---

Additive Energy of Truth Table Bounds ACC⁰ Circuit Size for Explicit Functions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Additive Combinatorics × ACC⁰ Circuit Size
- **Recorded**: 2026-05-12 15:52 UTC
- **Entry ID**: `1c8a17d67b25`

Statement

For any explicit Boolean function f: {0,1}^n → {0,1} in P, if its truth table's additive energy E(f) exceeds 2^{n/2}, then f cannot be computed by an ACC⁰ circuit of size < 2^{n/4}.

Rationale

High additive energy indicates structured correlations in the truth table that resist low-depth arithmetic circuit approximation, as ACC⁰ circuits lack the flexibility to encode such dependencies efficiently.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_91ef5bff.py", line 80, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_91ef5bff.py", line 63, in run_trial
    energy = truth_table_additive_energy(f)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_91ef5bff.py", line 19, in truth_table_a
    n = int(f.keys()[0].bit_length())
            ~~~~~~~~^^^
TypeError: 'dict_keys' object is not subscriptable
```

Judge reasoning

Test crashed due to TypeError in truth_table_additive_energy function. The error suggests incompatible dictionary key access methodology. | next: Fix the dictionary key subscripting error in truth_table_additive_energy by converting dict_keys to list first

---

Plethysm Multiplicity Inverse Proportional to Monotone Circuit Size for Permanent

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation Theory of Symmetric Groups (Plethysm Coefficients) × Monotone Circuit Size for Permanent Polynomial
- **Recorded**: 2026-05-12 16:08 UTC
- **Entry ID**: `2e8cd8937a3c`

Statement

For a random 3-SAT instance Φ with n variables, let μ(Φ) denote the plethysm coefficient of the symmetric power S^k(perm_n) decomposed into irreducible GL_n-modules. The monotone circuit size M(Φ) satisfies M(Φ) ≥ Ω(μ(Φ)^{-1} n^{2} log n).

Rationale

Plethysm coefficients encode symmetry constraints of tensor powers, which may impose structural limits on monotone circuits. Symmetric decomposition patterns could reveal inherent complexity barriers via representation-theoretic invariants.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.58s`

```
TRIAL: {'metric_name': 'monotone_circuit_size', 'metric_value': 29, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'monotone_circuit_size', 'metric_value': 47, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'monotone_circuit_size', 'metric_value': 43, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'monotone_circuit_size', 'metric_value': 32, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'monotone_circuit_size', 'metric_value': 18, 'instances_tested': 1, 'conjectu

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_49c604c9.py", line 106, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_49c604c9.py", line 45, in run_trial
    S_k_C = matrix_power(C, k)
            ^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_49c604c9.py", line 34, in matrix_power
    result = matrix_multiply(result, A)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_49c604c9.py", line 42, in matrix_multip
    result[i][j] += A[i][l] * B[l][j]
                              ~^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed during execution, preventing complete validation of the conjecture across all seeds. | next: Fix the matrix index error in test_49c604c9.py and re-run with sufficient seeds to meet the 80% support threshold.

Ehrhart Polynomial Coefficient Sum Bounded by Resolution Proof Size for 3-SAT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Ehrhart Theory × Resolution Proof Size for 3-SAT Instances
- **Recorded**: 2026-05-12 16:21 UTC
- **Entry ID**: `028df1797d70`

Statement

For a 3-SAT instance with n variables, let E(n) be the sum of absolute values of coefficients of its Ehrhart polynomial. The resolution proof size P(n) satisfies $E(n) \leq c * P(n)$ for some constant $c \geq 1$, and $P(n) \leq d * E(n)$ for another constant $d \geq 1$.

Rationale

Ehrhart polynomials encode geometric information about the integer point structure of solution spaces. By linking these coefficients to resolution proof size, we may uncover structural relationships between the geometry of the solution space and the complexity of proving unsatisfiability.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1c2aaa72.py", line 89, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1c2aaa72.py", line 49, in run_trial
    clauses = generate_3sat_instance(n)
              ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1c2aaa72.py", line 21, in generate_3sat
    clause = [random.choice(variables), -random.choice(variables)]
              ^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/random.py", line 348, in choice
    return seq[self._randbelow(len(seq))]
           ~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^
TypeError: 'set' object is not subscriptable
```

Judge reasoning

Test crashed due to TypeError from subscripting a set; no data collected to evaluate conjecture | next: Fix generate_3sat_instance to use list instead of set for variables

---

Noncommutative L^∞ Norm Lower Bounds Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Noncommutative L^p Geometry × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-12 17:13 UTC
- **Entry ID**: `a956dba16ef5`

Statement

For the DISJOINTNESS matrix $M \in \{0,1\}^{n\times n}$, the noncommutative L^∞ norm $||M||_{\infty} \geq c\sqrt{n}$ for some absolute constant $c > 0$. This norm is defined as the maximum singular value of the operator M acting on the space of matrices with trace-zero constraints.

Rationale

Noncommutative L^p norms capture operator-level structure of communication matrices, while DISJOINTNESS's Ω(√n) lower bound suggests deep algebraic constraints. The trace-zero constraint enforces nontrivial matrix-valued analysis, linking operator norms to communication complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.85s`

```
infinity_norm', 'metric_value': 5.991830142481126, 'instances_tested': 1, 'conjecture_holds': True,
TRIAL: {'metric_name': 'noncommutative_L_infinity_norm', 'metric_value': 5.96133670580432, 'instanc
TRIAL: {'metric_name': 'noncommutative_L_infinity_norm', 'metric_value': 6.6215302279845885, 'insta
TRIAL: {'metric_name': 'noncommutative_L_infinity_norm', 'metric_value': 6.14678256210607, 'instanc
TRIAL: {'metric_name': 'noncommutative_L_infinity_norm', 'metric_value': 6.0720069154488945, 'insta
TRIAL: {'metric_name': 'noncommutative_L_infinity_norm', 'metric_value': 6.0618401665591835, 'insta
TRIAL: {'metric_name': 'noncommutative_L_infinity_norm', 'metric_value': 6.15979419548317, 'instanc
TRIAL: {'metric_name': 'noncommutative_L_infinity_norm', 'metric_value': 5.889946751344519, 'instan
TRIAL: {'metric_name': 'noncommutative_L_infinity_norm', 'metric_value': 6.579239868327553, 'instan
TRIAL: {'metric_name': 'noncommutative_L_infinity_norm', 'metric_value': 6.456552365308594, 'instan
TRIAL: {'metric_name': 'noncommutative_L_infinity_norm', 'metric_value': 5.769737134003232, 'instan
TRIAL: {'metric_name': 'noncommutative_L_infinity_norm', 'metric_value': 6.215313582659221, 'instan
RESULT: SUPPORTED mean=6.141783163461413 std=0.2806895342283576 support_fraction=1.0
```

Judge reasoning

parse_fail | next: NONE

Quasigroup Idempotent Density Inversely Proportional to ACC⁰ Circuit Size for Parity-Insensitive Functions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Quasigroup Theory × ACC⁰ Circuit Size for Parity-Insensitive Functions
- **Recorded**: 2026-05-12 17:24 UTC
- **Entry ID**: `83d38b789ed6`

Statement

For a random quasigroup operation * on {0,1}^n, let I(∗) be the number of idempotent elements x with x*x=x. The ACC⁰ circuit size of the Boolean function f(x,y)=x*y is Θ(n² / I(∗)) for n ≥ 4.

Rationale

Quasigroups encode non-associative operations with Latin square properties, which may expose structural constraints on parity-insensitive functions. Idempotent density could quantify algebraic simplicity, inversely correlating with circuit complexity via Williams' algebraic geometry techniques.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.10s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing validation of support or falsification. | next: Increase timeout duration and re-run test with optimized parameters

---

Free Cumulant Sum Gap in Read-Twice BPs for IP_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Probability Theory × Read-Twice Branching Programs for IP_2
- **Recorded**: 2026-05-12 17:37 UTC
- **Entry ID**: `f36846330fde`

Statement

Let P be a read-twice BP for IP_2 with n variables. Define ρ(P) as the sum of free cumulants of the matrix M_P whose entries are the transition probabilities between BP states. Then ρ(P) = Ω(n) for all P, while ρ(Q) = O(log n) for any read-twice BP Q computing a function with poly(n) size circuits.

Rationale

Free cumulants capture noncommutative dependencies in BP transitions. IP_2's inherent symmetry forces exponential matrix rank, creating a free entropy gap that reflects the exponential BP size lower bound. This connects free probability's operator-algebraic structure to branching program complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `-9`, elapsed: `77.99s`

```
(no output)
```

Judge reasoning

Test crashed before producing data, preventing evaluation of support fraction or counterexamples. | next: Debug and rerun the test to collect empirical results

---

Tropical Variety Dimension Gap in Read-Twice vs Read-Once BPs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Tropical Geometry × Read-Twice Branching Programs
- **Recorded**: 2026-05-12 17:56 UTC
- **Entry ID**: `d35dbb9412cf`

Statement

For a read-twice branching program P over n variables, the dimension of its tropicalization (as a tropical variety) satisfies dim(trop(P)) ≥ n/2. For read-once BPs, dim(trop(P)) ≤ log n + 1.

Rationale

Tropical geometry captures combinatorial dependencies in BPs via tropical varieties. Read-twice BPs introduce multiplicative interactions that expand tropical dimension linearly, while read-once BPs have additive structure limiting dimension to logarithmic growth.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fd822c45.py", line 101, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fd822c45.py", line 82, in run_trial
    read_once_dim = tropical_dimension(tropical_matrix(read_once_bp))
                    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fd822c45.py", line 44, in tropical_dime
    if all(abs(matrix[j][i]) == float('inf') for j in range(n)):
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fd822c45.py", line 44, in <genexpr>
    if all(abs(matrix[j][i]) == float('inf') for j in range(n)):
               ~~~~~~~~~^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError during tropical dimension calculation, preventing validation of conjecture | next: Debug tropical_matrix implementation for read-once BPs and re-run tests

---

Real Rank of SOS Moment Matrices for Max-CUT Bounded by Degree

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × SOS Degree for Max-CUT
- **Recorded**: 2026-05-12 18:08 UTC
- **Entry ID**: `3fc630b9a37e`

Statement

For a random graph G(n, 1/2) with n ≤ 40, the real rank of its degree-d SOS moment matrix M_d satisfies rank(M_d) ≤ d^2. If this fails, then the integrality gap of the d-round Lasserre hierarchy for Max-CUT on G exceeds 0.878.

Rationale

The real rank of moment matrices encodes the algebraic structure of polynomial certificates. By linking it to SOS degree, we expose how geometric constraints on positive semidefiniteness constrain approximation ratios, potentially revealing barriers to higher-degree hierarchies.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_40a0efbf.py", line 91, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_40a0efbf.py", line 24, in run_trial
    M_d = degree_sos_moment_matrix(G, d)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_40a0efbf.py", line 63, in degree_sos_mo
    M_d[n + i][n + j] += math.comb(k, 2)
    ~~~~~~~~~~^^^^^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed before collecting data, preventing verification of conjecture | next: Fix the IndexError in degree_sos_moment_matrix implementation and re-run experiments

---

Submodular Rank Gap in Monotone DNF for k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid Theory (Polymatroid Expansion) × Monotone Circuit Size for k-CLIQUE
- **Recorded**: 2026-05-12 18:22 UTC
- **Entry ID**: `6ae57c36380b`

Statement

For any k-CLIQUE instance with $n \geq 40$, the submodular rank of its characteristic set system is $\Omega(n)$, while any monotone DNF formula of size $O(\text{poly}(n))$ has submodular rank $O(\log n)$.

Rationale

Polymatroid rank captures the combinatorial structure of k-CLIQUE's hypergraph, and its submodular gap may reflect the inherent complexity of monotone computation. The conjecture links matroid-theoretic invariants to circuit size via the polymatroid expansion of the problem's incidence matrix.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e89e0fcc.py", line 120, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e89e0fcc.py", line 102, in run_trial
    rank = submodular_rank(A)
           ^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e89e0fcc.py", line 55, in submodular_ra
    A[j][i] -= A[j][max_col] * A[max_col][i] / A[max_col][max_col]
               ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^~~~~~~~~~~~~~~~~~~~
ZeroDivisionError: division by zero
```

Judge reasoning

Test crashed with ZeroDivisionError, preventing data collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Debug the ZeroDivisionError in submodular_rank computation by adding numerical stability (e.g., epsilon regularization) and retesting

---

Euler Characteristic of Communication Graph Equals Deterministic Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic topology (Euler characteristic of simplicial complexes) × Deterministic communication complexity of Boolean functions
- **Recorded**: 2026-05-12 19:26 UTC
- **Entry ID**: `1d0355b7e76a`

Statement

For any Boolean function f, the Euler characteristic of the clique complex of its communication graph is equal to the deterministic communication complexity of f.

Rationale

The Euler characteristic captures topological invariants that may reflect the structural complexity of the communication graph, which could directly relate to the required communication. This bridge might expose how geometric features of the interaction graph correlate with communication cost.

Novelty

- Judge: `NOVEL` over 6 arXiv hits

Top hits consulted: - [1511.07912v3] The Thom-Sebastiani theorem for the Euler characteristic of cyclic L-infinity algebras - [2509.13173v1] Euler's explorations of extremal ellipses - [2003.09703v1] Variance function of boolean additive convolution - [1112.4523v1] Complexity and Algorithms for Euler Characteristic of Simplicial Complexes - [1808.02841v1] On divergent Series

Empirical Test

- exit code: `-9`, elapsed: `95.36s`

```
(no output)
```

Judge reasoning

Test crashed before producing data, preventing evaluation of support fraction or counterexamples. | next: Re-run test with enhanced logging to diagnose crash cause and validate conjecture

---

Completely Bounded Norm Gap in Read-Twice BPs for IP_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Operator Space Theory × Read-Twice Branching Programs
- **Recorded**: 2026-05-12 19:34 UTC
- **Entry ID**: `e1525ec3333f`

Statement

Let M be the transition matrix of a read-twice BP P. The completely bounded norm ||M||*{cb} satisfies ||M||*{cb} = O(log size(P)) for all P, but ||M||_{cb} = Ω(n) for any BP computing IP_2 mod 2.

Rationale

Operator space norms capture non-commutative structure of matrices, which may expose inherent complexity in read-twice BPs. IP_2's exponential lower bound for read-once BPs suggests its read-twice representation must exhibit nontrivial operator space properties.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.34s`

```
TRIAL: {'metric_name': 'cb_norm', 'metric_value': 4.031266373900718e-25, 'instances_tested': 1, 'con
25 < 1.65'}
TRIAL: {'metric_name': 'cb_norm', 'metric_value': 5.271483498975786e-26, 'instances_tested': 1, 'con
26 < 1.15'}

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9ab3931c.py", line 140, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9ab3931c.py", line 124, in run_trial
    cb_norm = spectral_norm(M)
              ^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9ab3931c.py", line 117, in spectral_nor
   return max(abs(eigenvalue) for eigenvalue in solve_linear_system(matrix_inverse(B), [sum(row[i] fo
                                                                     ^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9ab3931c.py", line 92, in matrix_invers
    A[k][j] /= pivot
ZeroDivisionError: float division by zero
```

Judge reasoning

Test crashed with ZeroDivisionError, invalidating all trial results. The counterexamples shown may be artifacts of failed computation. | next: Fix the matrix inversion division-by-zero error in spectral_norm implementation

---

Non-Abelian Fourier Coefficient Spread Inversely Proportional to Resolution Proof Length for Tseitin Formulas

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Non-Abelian Harmonic Analysis × Resolution Proof Length for Tseitin Formulas
- **Recorded**: 2026-05-12 19:42 UTC
- **Entry ID**: `2bf1ffaff7e5`

Statement

For a Tseitin formula Φ over a connected planar graph G, the spread of non-abelian Fourier coefficients (max_k |F(k)| - min_k |F(k)|) of its truth table is inversely proportional to the minimal resolution proof length of Φ, with proportionality constant $\Theta(\sqrt{n})$.

Rationale

Non-abelian Fourier analysis captures symmetries in boolean functions' truth tables, while Tseitin formulas' resolution complexity is tied to graph structure. The conjecture links algebraic structure (non-abelian Fourier coefficients) to combinatorial proof complexity, potentially revealing hidden invariants in planar satisfiability.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0e8e9054.py", line 117, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0e8e9054.py", line 85, in run_trial
    graph = generate_planar_graph(n)
            ^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0e8e9054.py", line 27, in generate_plan
    raise ValueError("Unsupported graph size for this simple generator")
ValueError: Unsupported graph size for this simple generator
```

Judge reasoning

Test crashed due to unsupported graph size, preventing data collection to verify proportionality claim | next: Implement a more robust planar graph generator supporting larger n values

---

Secant Rank of Disjointness Communication Matrix Lower-Bounds Randomized Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Geometry (Secant Varieties of Veronese Embeddings) × Randomized Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-12 20:10 UTC
- **Entry ID**: `c4fba4dde224`

Statement

For the DISJOINTNESS matrix M_n ∈ {0,1}^{n×n}, its secant rank sr(M_n) satisfies sr(M_n) ≥ Ω(n). This follows from the fact that the Veronese variety V_d ⊊ P^N (d=2) has secant variety σ(V_d) of dimension 2d−1, and M_n lies in σ(V_d) only if n ≤ d^{1/2} + 1.

Rationale

Secant rank captures the minimal number of rank-1 tensors needed to approximate a matrix, mirroring the communication complexity of DISJOINTNESS which requires Ω(n) bits. The Veronese embedding's secant variety dimension provides a geometric obstruction to low-rank approximations of the communication matrix.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds':
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds':
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds':
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds':
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds':

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_427ba9ca.py", line 90, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_427ba9ca.py", line 71, in run_trial
    sr_M_n = secant_rank(M_n)
             ^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_427ba9ca.py", line 63, in secant_rank
    C = matrix_multiply(A, transpose(B))
                           ^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_427ba9ca.py", line 28, in transpose
    return [[A[j][i] for j in range(len(A))] for i in range(len(A[0]))]
                                                                ~^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed during execution, preventing reliable evaluation of conjecture validity. | next: Implement error handling for matrix transposition and re-run tests with validated matrix dimensions

Schur Coefficient Sum Ratio in Symmetric Powers of Permanent vs Determinant

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl Duality × Circuit Size for Symmetric Polynomials
- **Recorded**: 2026-05-12 20:40 UTC
- **Entry ID**: `d8795d1b709f`

Statement

For each n ≥ 2, let S_n be the sum of Schur coefficients c_λ(perm_n) for all partitions λ of n with λ_1 ≤ n^{1/2}. Then, S_n / S'_n > 2^{n/2}, where S'_n is the analogous sum for det_n.

Rationale

The permanent's symmetric structure under Schur-Weyl duality leads to a richer decomposition in symmetric powers, which could imply higher circuit complexity. The exponential separation suggests a fundamental asymmetry in their representation-theoretic properties.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_102f931e.py", line 105, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_102f931e.py", line 80, in run_trial
   S_n = sum(schur_coefficient(perm_matrix, partition) for partition in partitions(n) if max(partitio
                                                                        ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_102f931e.py", line 26, in partitions
    return partitions_recursive(n)
           ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_102f931e.py", line 22, in partitions_re
    for i in range(1, min(n + 1, max_partition + 1)):
                                 ~~~~~~~~~~~~~~^~~
TypeError: unsupported operand type(s) for +: 'NoneType' and 'int'
```

Judge reasoning

Test crashed due to TypeError in partitions_recursive function. The error suggests a missing initialization for max_partition variable. | next: Debug partitions_recursive function to handle max_partition initialization and fix the NoneType + int operation

---

Hook-Length Ratio Bounds Monotone Circuit Size for Permanent

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Combinatorics (Hook-Length Formula) × Monotone Circuit Size for Permanent
- **Recorded**: 2026-05-12 21:10 UTC

- **Entry ID**: `c99c02cb9842`

Statement

For every n ≥ 1, the ratio of standard Young tableaux counts for the permanent's Young diagram λ_n to the determinant's μ_n satisfies |SYT(λ_n)| / |SYT(μ_n)| ≥ 2^{n/2}. This ratio equals the dimension of the symmetric power of the defining representation divided by the exterior power dimension.

Rationale

The hook-length formula quantifies representation dimensions tied to permutation symmetries. Permanent's symmetric structure forces exponentially larger tableau counts than determinant's alternating structure, suggesting inherent complexity in monotone computation.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5fc5efa6.py", line 79, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5fc5efa6.py", line 55, in run_trial
    ratio = SYT_λ_n / SYT_μ_n
            ~~~~~~~~^~~~~~~~~
ZeroDivisionError: float division by zero
```

Judge reasoning

Test crashed with division by zero, preventing evaluation of the ratio. The error suggests a potential flaw in SYT_μ_n calculation or invalid input for some n. | next: Debug the SYT_μ_n computation to verify if μ_n's Young diagram is valid for all n, and check if division by zero occurs for specific cases.

Irreducible Component Count Inversely Proportional to SOS Refutation Degree for 3-SAT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Geometry over Finite Fields × SOS Refutation Degree for 3-SAT
- **Recorded**: 2026-05-12 22:03 UTC
- **Entry ID**: `1f09f4ec8317`

Statement

For a random 3-SAT instance with n variables, the number of irreducible components in the solution variety over GF(2) is Θ(1/(d^2)) where d is the SOS refutation degree. All unsatisfiable instances with ≤40 variables satisfy this bound.

Rationale

The algebraic structure of the solution space (number of irreducible components) directly influences SOS refutation complexity. Higher component count implies more intricate geometry, requiring higher-degree certificates to prove unsatisfiability.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.69s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing data; insufficient evidence to meet support fraction criterion | next: Run test with extended timeout and optimized parameters for larger instances

---

Gowers Norm Inverse Proportional to ACC^0 Circuit Size for Explicit Functions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Additive Combinatorics × ACC^0 Circuit Size
- **Recorded**: 2026-05-12 22:47 UTC
- **Entry ID**: `331ee94784d2`

Statement

For a Boolean function f: {0,1}^n → {0,1}, the Gowers norm of order 2 (||f||*{U^2}) is inversely proportional to the minimal ACC^0 circuit size computing f. Specifically, ||f*||{U^2} ≤ 1/√(S(f)) where S(f) is the minimal ACC^0 circuit size.

Rationale

Gowers norms quantify pseudorandomness, which may correlate with circuit complexity. ACC^0 circuits are limited in their ability to compute functions with high pseudorandomness, creating a potential inverse relationship between these measures.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.34s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Run test with stricter time limits and increased seed diversity to resolve ambiguity

---

Irreducible Component Count of IP_2 BP Variety Bounds Read-Twice Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Geometry (Irreducible Components of Zero Sets) × Read-Twice Branching Programs for IP_2
- **Recorded**: 2026-05-12 22:54 UTC
- **Entry ID**: `dd74abbacec1`

Statement

For any read-twice BP P computing IP_2 on n variables, the number of irreducible components of the variety defined by P's constraints is Ω(n). For read-once BPs, this count is O(log n).

Rationale

The algebraic structure of IP_2's constraints forces read-twice BPs to encode higher-dimensional varieties, while read-once BPs capture lower-dimensional structures. This geometric disparity mirrors the exponential size gap between BP models.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.03s`

```
, 'counterexample': 'mapping_undefined'}
TRIAL: {'metric_name': 'irreducible_components', 'metric_value': 2, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'irreducible_components', 'metric_value': 2, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'irreducible_components', 'metric_value': 2, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'irreducible_components', 'metric_value': 2, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'irreducible_components', 'metric_value': 2, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'irreducible_components', 'metric_value': 2, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'irreducible_components', 'metric_value': 2, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'irreducible_components', 'metric_value': 2, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'irreducible_components', 'metric_value': 2, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'irreducible_components', 'metric_value': 2, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'irreducible_components', 'metric_value': 2, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'irreducible_components', 'metric_value': 2, 'instances_tested': 1, 'conjectu
RESULT: FALSIFIED counterexample="mapping_undefined" first_failing_seed=11
```

Judge reasoning

Single-instance testing cannot validate Ω(n) claims; metric value is constant (2) across all trials, failing to distinguish read-twice vs read-once cases. Counterexample 'mapping_undefined' suggests computational errors in variety calculation. | next: Test with n ≥ 1000 and validate variety computation using certified algebraic geometry tools

---

Irreducible Component Count Inversely Proportional to ACC⁰ Circuit Size for Sipser-like Functions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Geometry over Finite Fields × ACC⁰ Circuit Size for Explicit Functions
- **Recorded**: 2026-05-12 23:21 UTC

- **Entry ID**: `f5574a6cc9ec`

Statement

For a random SAT instance with n variables, the number of irreducible components of its defining polynomial over GF(2) is at least $\Omega(2\hat{}\{n/2\})$ if the function cannot be computed by an ACC^0 circuit of size $o(2\hat{}\{n/2\})$.

Rationale

Algebraic geometry provides a framework to analyze the structure of Boolean functions via their polynomial representations. The irreducible components of the defining polynomial encode the function's geometric complexity, which may correlate with circuit size barriers in ACC^0.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.23s`

```
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

Judge reasoning

Test crashed before producing conclusive data; pre-registered support condition cannot be verified | next: Run test with increased time/memory limits and verify output format

---

Free Cumulant Gap in Read-Twice BPs for 3-SAT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Probability Theory × Read-Twice Branching Programs
- **Recorded**: 2026-05-12 23:54 UTC
- **Entry ID**: `8818af330631`

Statement

For a read-twice BP computing a 3-SAT instance with n variables, the maximum free cumulant of its truth table is $O(\log n)$. For the IP_2 function (inner product mod 2), which requires exponential size read-twice BPs, the maximum free cumulant is $\Omega(n)$.

Rationale

Free cumulants capture non-commutative independence structures in random matrices, which may reveal hidden algebraic constraints in BP computation. The IP_2 function's exponential lower bound suggests its cumulant growth diverges from read-twice BPs' logarithmic capacity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.34s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support/falsification criteria. | next: Run test with extended timeout and higher resource allocation